Data retention and deletion

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Data retention and deletion before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sandboxing and resource limits

Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

External side effects and recovery

Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Planning and action selection

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

Toy policy retains session content for 7 days and deletes it when age is at least 7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

Practise with fading help

> Complete the supported Data retention and deletion trace and record the first step you can explain without help.

Use W2: Trace the worked case for Data retention and deletion. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Data retention and deletion explanation against the supplied facts before discussing it.

Use the worked case for Data retention and deletion. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Fictional retention policy deletes records at age >=4 days. X is3 days, Y4, Z5. A copy of Y remains in a temporary log after its main record is deleted. Evaluate compliance.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Data retention and deletion facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied Data retention and deletion facts and keep your reasoning.

Record ages are 6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Data retention and deletion facts and keep your reasoning.

A report omits this qualification: This fictional retention exercise is not a statement of legal retention requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Data retention and deletion facts and keep your reasoning.

Retention is5 days, inclusive **deletion boundary**. Ages 4, 5, 6; a log stores complete content. Trace required action.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Data retention and deletion facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Data retention and deletion to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled data retention and deletion **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of data retention and deletion.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

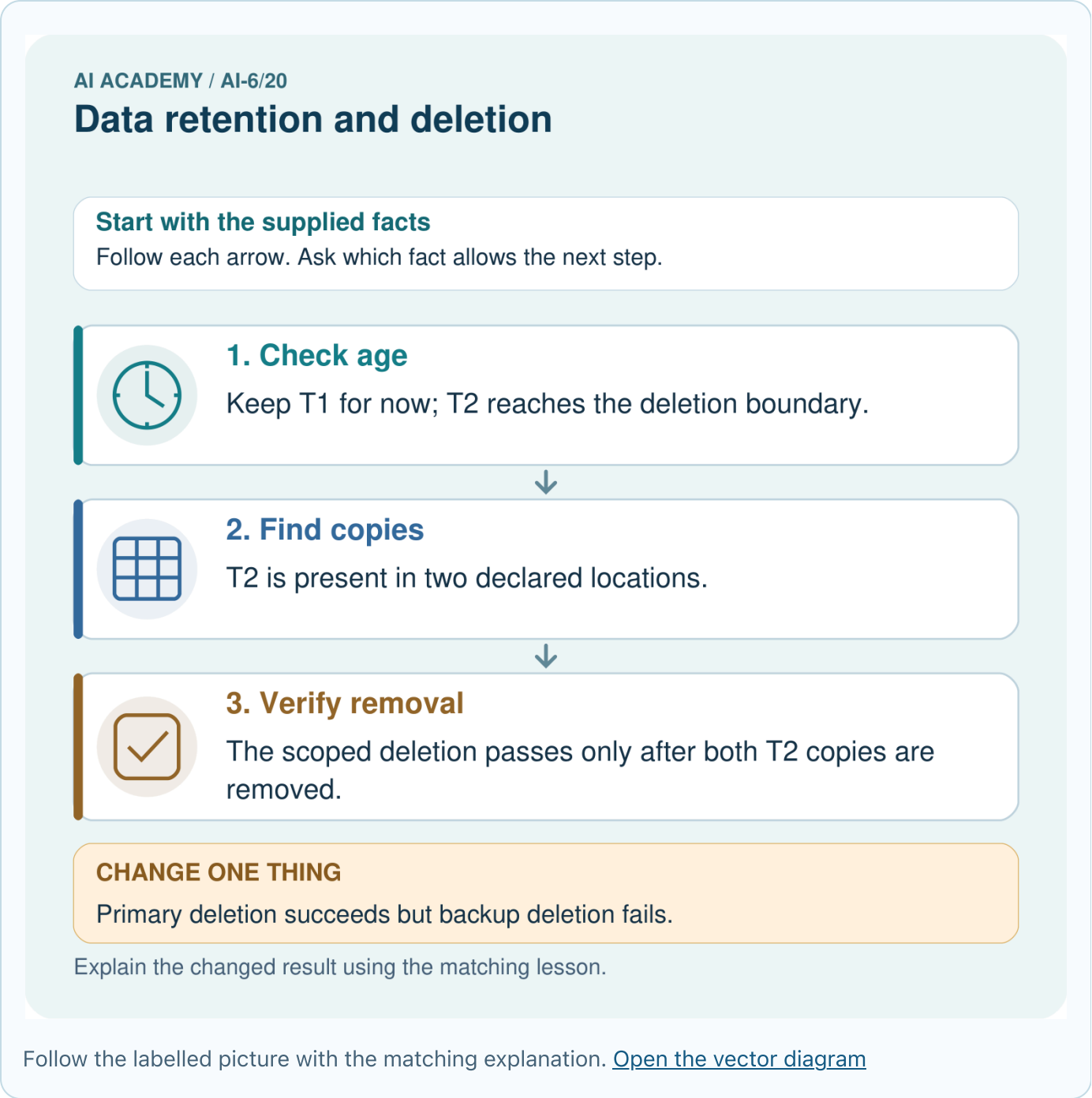
Pictorial case practice: Data retention and deletion

> Delete every listed copy. A fictional policy deletes transcripts at age seven days or older,

> including listed backups. Explain the labelled step before checking the changed condition.

Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.

T1 is six days old. T2 is seven days old with one backup. Scope lists primary storage and that backup only.



Follow the labelled picture with the matching explanation. Open the vector diagram

Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Project portfolio milestone

Keep this week’s input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The scoped deletion passes only after both T2 copies are removed.

1. Check age

2. Find copies

3. Verify removal

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Verify removal The scoped deletion passes only after both T2 copies are removed. Retention requires an age rule and an inventory.

Deleting the visible copy alone does not demonstrate removal of listed backups.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Primary deletion succeeds but backup deletion fails. Which conclusion is supported by the supplied facts?

1. Record incomplete deletion and follow the retry or escalation procedure.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Record incomplete deletion and follow the retry or escalation procedure. One required copy remains in the declared scope.

Repair a missing building block

Review Sandboxing and resource limits (ai-6/19). Goal: Explain the **mechanism** and terminology of sandboxing and resource limits. Short practice: Check A against both resource limits and B against network isolation.

Why must A stop after three seconds even if its **output** remains only 2 MB?. Return to this week and explain what you changed.

Review External side effects and recovery (ai-6/17). Goal: Explain the **mechanism** and terminology of external side effects and recovery. Short practice: Distinguish a missing printer acknowledgement from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?. Return to this week and explain what you changed.

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

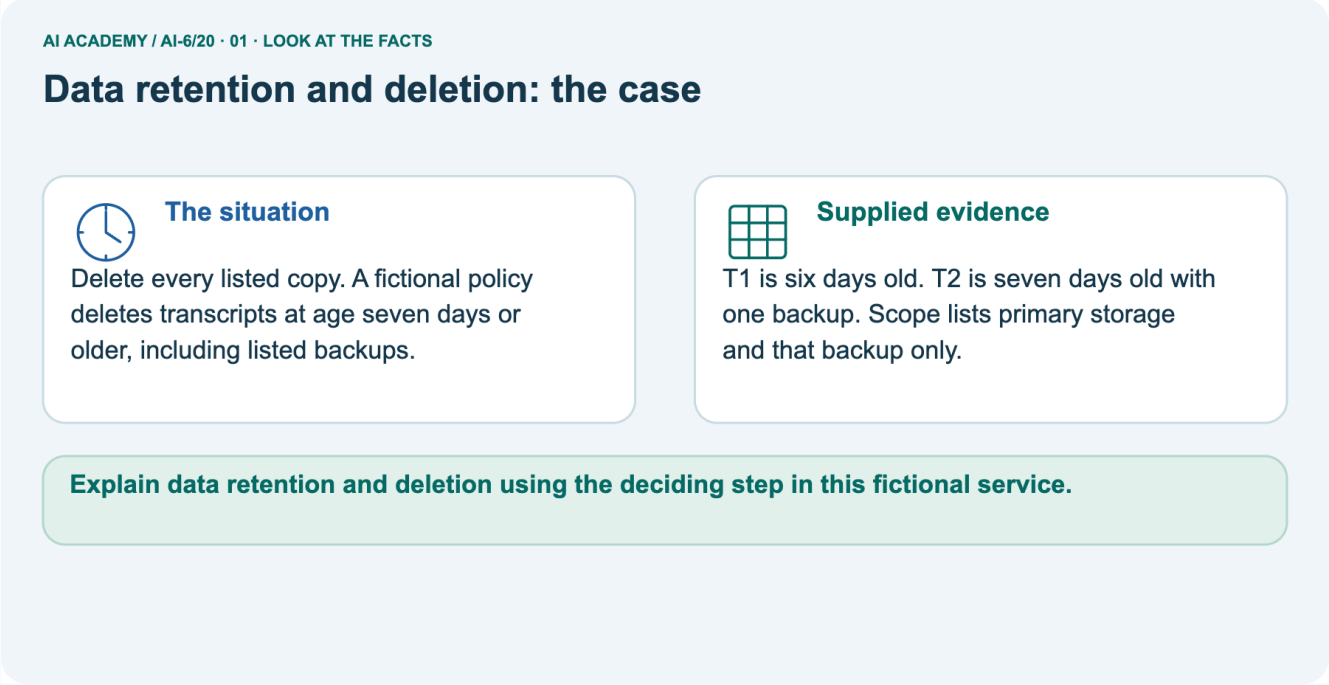
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Data retention and deletion picture from its labels, then record a relevant limit.

01. Data retention and deletion: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

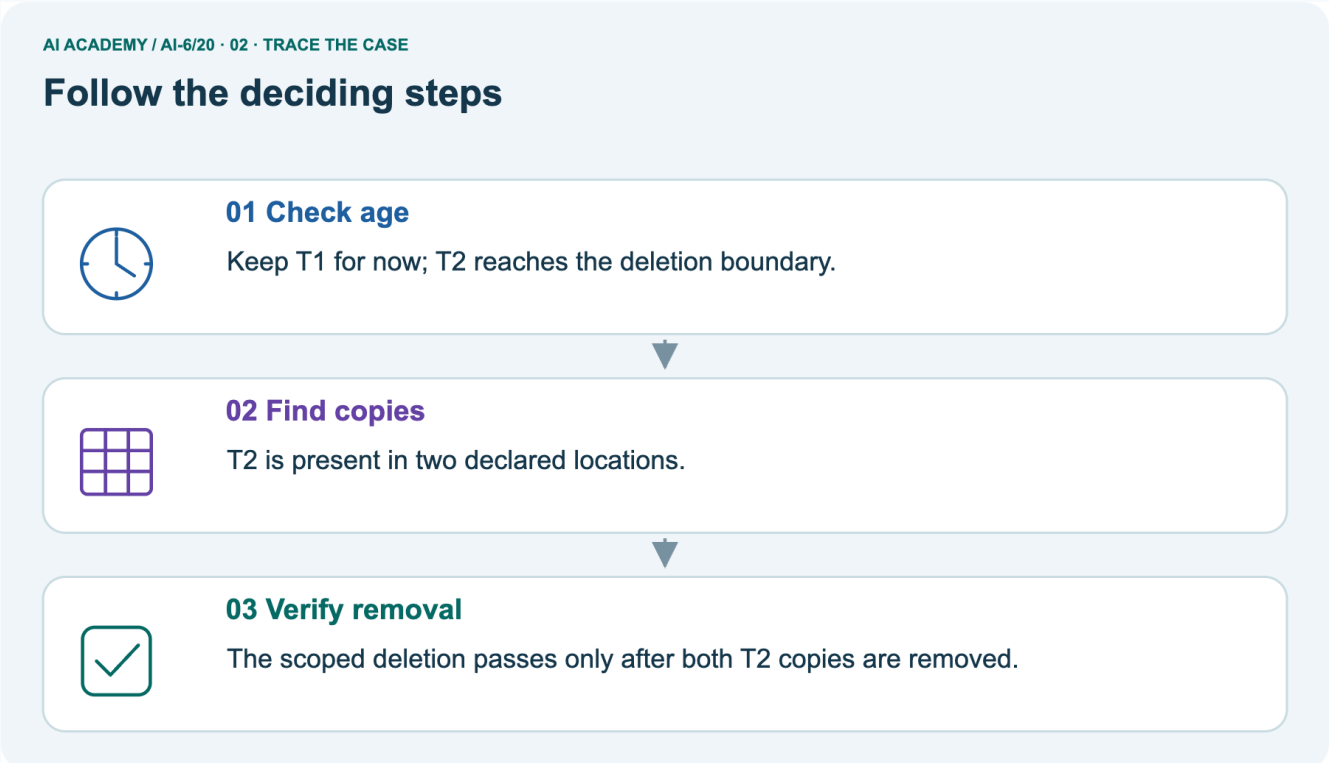
Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups. Supplied facts: T1 is six days old. T2 is seven days old with one backup.

Scope lists primary storage and that backup only.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



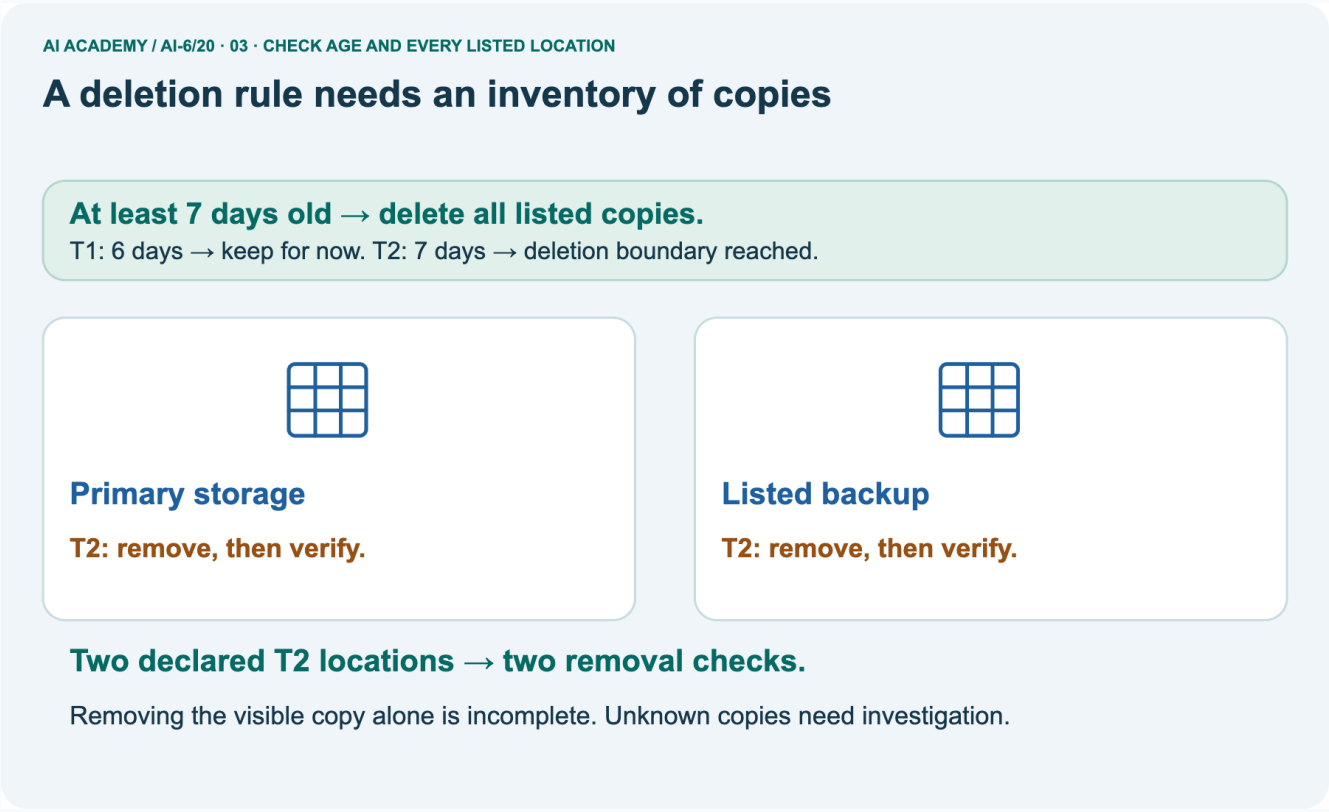
Follow the labelled picture with the matching explanation. Open the vector diagram

Check age: Keep T1 for now; T2 reaches the **deletion boundary**. → Find copies: T2 is present in two declared locations. → Verify removal: The scoped deletion passes only after both T2 copies are removed.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A deletion rule needs an inventory of copies



Follow the labelled picture with the matching explanation. Open the vector diagram

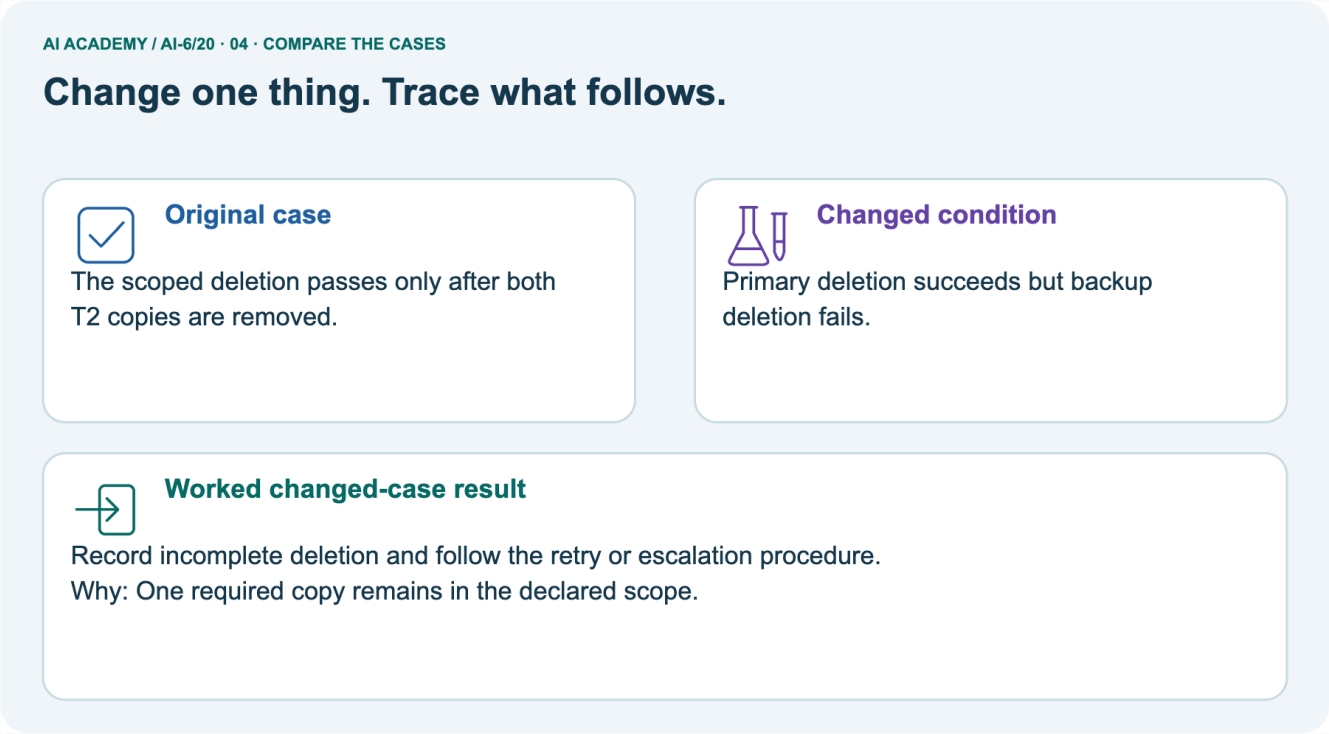
The toy retention boundary is seven days or older. Keep six-day T1 for now. Seven-day T2 appears in primary storage and one listed backup; both locations need removal verification.

This diagram specifies required checks, not an actual deletion run.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

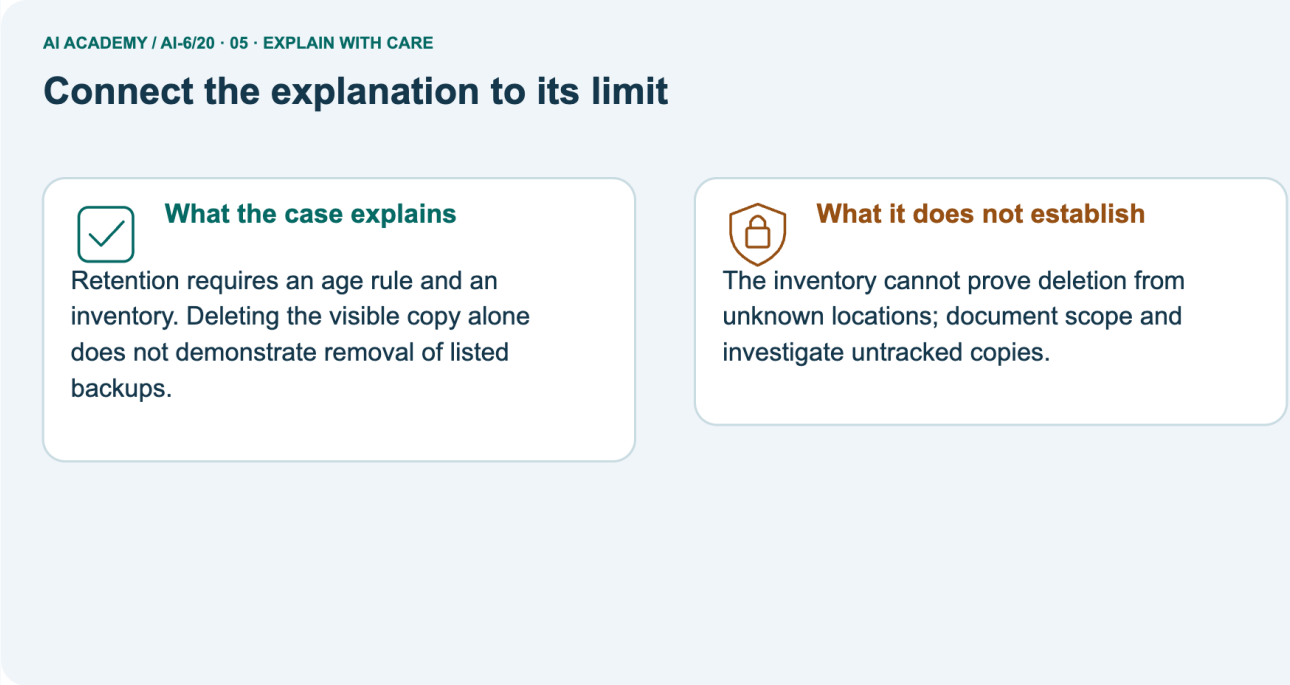
Original case: The scoped deletion passes only after both T2 copies are removed. Changed condition: Primary deletion succeeds but backup deletion fails.

Expected result: Record incomplete deletion and follow the retry or escalation procedure. Reason: One required copy remains in the declared scope.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

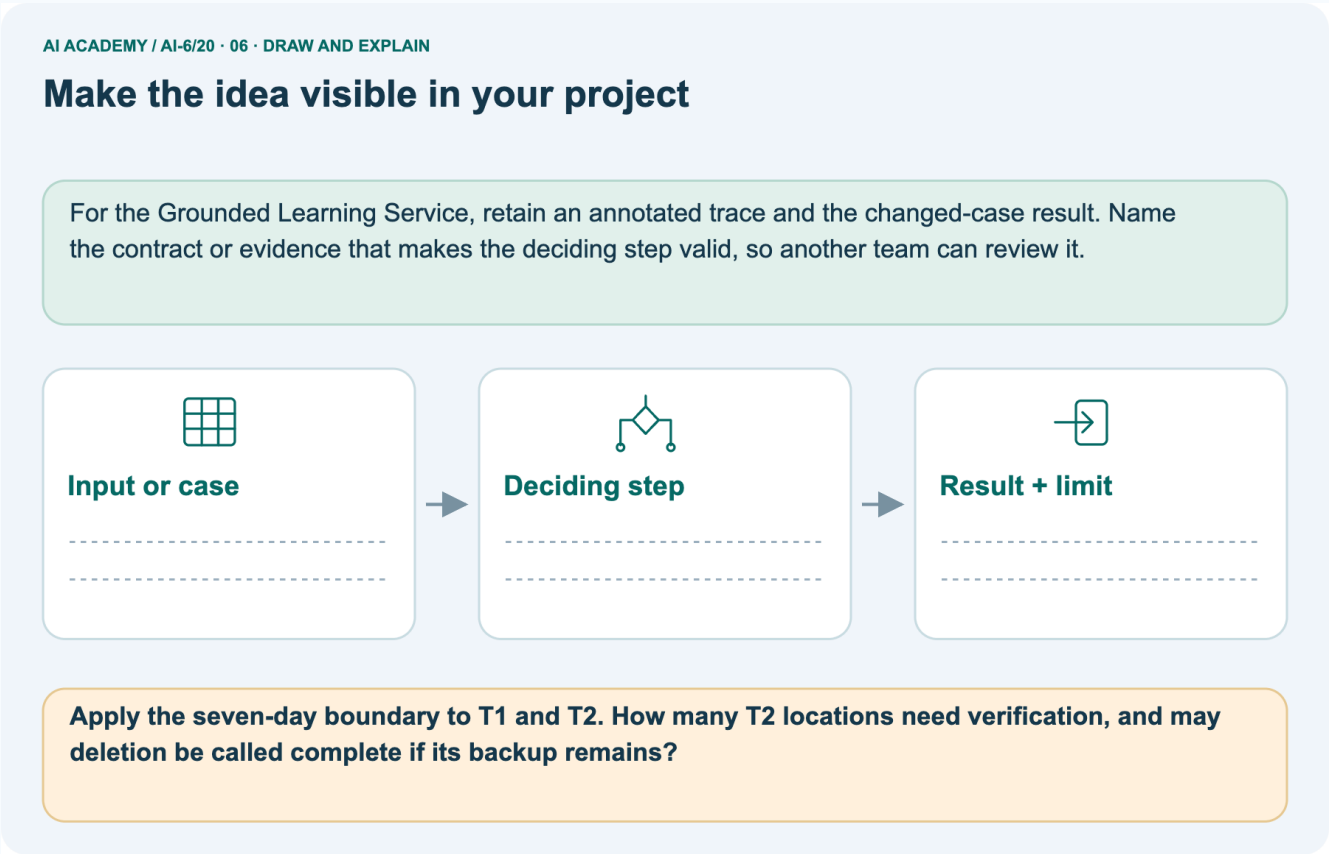
Retention requires an age rule and an inventory. Deleting the visible copy alone does not demonstrate removal of listed backups.

Limit: The inventory cannot prove deletion from unknown locations; document scope and investigate untracked copies.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.